

Quentin MELOTTE

Apprentice
Research Engineer

About Me

Hi, I'm Quentin – an engineering student passionate about understanding the inner workings of electronics and computer science. I'm driven by research and particularly interested in embedded systems, with a focus on aerospace and robotics.

Contact

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🌐 <https://nayggets.github.io/>

Skills

- Computer Architecture Knowledge
 - Von Neumann/Harvard model
 - Memory hierarchy
 - MMU/DMA
- Compiler Conception and Development
 - Parser/Lexer (ANTLR4, recursive descent in C)
 - LLVM (knowledge only)
 - AST Visitor development
- Chip Design and Conception
 - HDL (VHDL, Verilog)
 - Emulation/simulation
 - CPU conception
- Embedded system development
 - STM32/STM8
 - Arduino
 - Raspberry
- Programming
 - Object oriented
 - Imperatif
 - Procedural
 - Generic
- Assembly (x86, RISCv, ARM)
 - Stack management
 - Code Optimisation
- More skill and proof of skill on my webSite

Experiences

Apprentice Research Engineer – Compiler Technology 2024–now
CEA List – Grenoble

- Supporting a research fellow on code optimization for High utility algorithm
- Developing a pedagogical language to help students understand programming language concepts more easily
- Contributing to a research publication in the field of compiler optimizations
- Learning and applying tools such as ANTLR and LLVM
- Exploring the HybroGen infrastructure and designing new compiler optimizations

Apprentice Developer and Maintainer – Video Hypervisor 2023–2024
ActiaRailway – Montpellier

- Contributed to feature development and bug fixing on a video and sound hypervisor
- Performed testing and validation on embedded systems (cameras, sensors)
- Maintained software security components: cryptographic libraries, internal tools, and certificates
- Conducted ONVIF compliance testing
- Initiated development of an MP4-to-JPEG video format converter

Intern – Network Security Protocol Development April – June 2023
ActiaTelecom – Montpellier

- Developed and integrated C/C++ implementations of RTSPS and SRTP to secure RTSP/RTP video streams
- Implemented the MIKEY key exchange protocol
- Embedded the complete solution into the company's hypervisor

Education

Bachelor of Computer Science 2021–2024
Montpellier University

During my Bachelor's degree specialized in cybersecurity, networking, and low-level systems. I gained experience in both high- and low-level programming, and explored key topics such as compiler design, network architecture, cryptography, scripting, algorithms, and performance-oriented programming.

Engineering degree Specialized in Cybersecure Intelligent Systems 2024–2027
Valence Grenoble INP – ESISAR

The main goal is to strengthen my foundation in electronics, hardware design, RF systems, computer architecture, and embedded systems, while refining the skills acquired during my Bachelor's. (See ESISAR program [here](#).)

Language

- English (B2, TOEIC)
- French (native)